

MANJUNATH K G

+91 8431665322 | manjunathkg4433@gmail.com | Bengaluru, India | [GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [LeetCode](#) | [TryHackMe](#)

EXPERIENCE

AI/ML Trainee Engineer

Sep 2025 – Present

Ampcus Cyber, Bengaluru

- Architected a production-grade autonomous pentesting SaaS with **60+ specialized AI agents** covering web, API, Android, source-code, and network surfaces; orchestrated via **Claude Agent SDK & LiteLLM** on a FastAPI/Next.js/PostgreSQL/Redis stack with Temporal workflows and public REST APIs for enterprise integrations
- Engineered a **phased orchestration pipeline** executing 30+ AI agents in parallel waves with checkpointing, fault tolerance, and workflow recovery via Temporal, reducing security assessment turnaround time from **3-5 days of manual effort to under 4 hours** while expanding coverage across OWASP Top 10 and OWASP WSTG testing categories.
- Developed the **automated report-generation module**, reducing report turnaround from **6 hours to 10 minutes (~97%)** and designing it for scale across concurrent client engagements.
- Owned the platform **end-to-end** — **feature development, deployment to test servers, and production incident response** — resolving client escalations ahead of deadline
- Built an **AWS MCP server exposing 70+ tools** for automated evidence fetching, integrating cloud data directly into the company's GRC application.
- Extended GRC evidence collection across **AWS, Azure, Jira, GitHub and GCP** via MCP agents, automating cloud-infrastructure compliance evidence gathering
- Drove **LLM output quality and cost optimization** — system prompts, few-shot templates, and chain-of-thought scaffolds — and **reduced token usage by 30%** through prompt tuning, response caching, and model-tier selection.
- Built a data-driven **RAG evaluation and monitoring framework** using RAGAS, enabling continuous assessment of retrieval and generation quality, improving answer faithfulness and relevancy by **30%** while preventing performance regressions during model and prompt updates.
- Partnered with **red-team, blue-team and GRC teams** to translate security insights into product features, improving application usability and adoption.
- Mentored colleagues on practical AI-tool adoption, running enablement sessions to integrate AI into daily workflows.
- Containerized all services with Docker** and **followed spec-driven development (SpecKit)** plus secure-coding principles for maintainable, auditable delivery.

TECHNICAL SKILLS

Agentic AI & LLM Engineering: Multi-Agent Systems, LLM Orchestration, Agentic Workflows, Prompt Engineering, Fine-Tuning, LLM Evaluation (RAGAS), Claude Agent SDK, LangChain, LangGraph, CrewAI, Google ADK, n8n

Agent Architecture & Patterns: ReAct, Plan-and-Execute, Reflection, Supervisor/Router patterns, Tool Calling, Human-in-the-Loop, Memory-Driven Agents, Planning & Reasoning, Workflow Orchestration

RAG & Knowledge Systems: Retrieval-Augmented Generation (RAG), Agentic RAG, Hybrid Search, Embeddings, Vector Databases (Pinecone, Chroma, pgvector), Knowledge Graphs, Short/Long-Term Memory, Context Management

MCP & Tool Integration: Model Context Protocol (MCP), MCP Server/Client Development, Custom Tool Development, Function Calling, REST/API Integrations, Agent-to-Tool Communication

LLM Platforms & Frameworks: Anthropic Claude API, OpenAI API, Google Gemini API, Hugging Face, Transformers, Streamlit

Backend & Infrastructure: FastAPI, REST APIs, Next.js, React, TypeScript, Docker, Temporal, Redis, PostgreSQL, MongoDB, MySQL, AWS / Azure / GCP

ML & Data Science: Scikit-learn, NumPy, Pandas, Matplotlib, Feature Engineering, Model Training & Evaluation, Deep Learning Fundamentals

Observability & Practices: Prometheus, Grafana, CI/CD, Spec-Driven Development (SpecKit), Secure Coding

Security: CEH v13, OWASP Top 10, MITRE ATT&CK, Burp Suite, Nmap, Nessus, Playwright

Tools: Claude Code, Cursor IDE, Jupyter, Postman, Git, Linux

EDUCATION

Bachelor of Computer Applications (BCA) · ASC Degree College, Bangalore University · CGPA: 8.3 / 10

2022 – 2025

CERTIFICATIONS & ACHIEVEMENTS

- Certified Ethical Hacker (CEH v13 with AI), EC-Council** · Credential ID: ECC5074128936
- Winner — Internal Company Hackathon (GRACEATHON):** Built a real-time infrastructure monitoring and observability agent (Python, FastAPI, Prometheus) tracking system health across production servers.
- AI Mastery with Machine Learning** — Bluetick AI Academy